Exercises
Probabilistic Artificial Intelligence
Fall 2025

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For questions, please refer to Moodle.
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Homework 1
(Probability, Bayesian Linear Regression)

General Instructions

- Submission of solutions is not mandatory but solving the exercises are highly recommended. The master solution will be released after the exercise deadline.
- Part of the exercises are available on Moodle as a quiz. These problems are marked with [☑].

Exercise 0: Warm-Up

Throughout this homework p denotes a probability mass function or density, and $\mathcal{N}(\mu, \Sigma)$ a Gaussian with mean μ and covariance Σ.

(a) Let W, Z be continuous random variables and suppose the conditional density of W does not depend on the value of Z, i.e. $p(W \mid Z = z) = q(W)$ for every $z \in \mathbb{R}$ with $p_Z(z) > 0$. Show that $p(W) = q(W)$.

Solution:

$$\begin{aligned} p(w) &= \int p(w, z)\, dz && \text{sum rule} \\ &= \int p(w \mid z)\, p(z)\, dz && \text{product rule} \\ &= q(w) \int p(z)\, dz = q(w). && p(w \mid z) = q(w)\ \forall z,\ \textstyle\int p(z)\, dz = 1 \end{aligned}$$

(b) Let f be a density on $\mathbb{R}$ with $f(x) \propto \exp\left(-\frac{1}{2}ax^2 + bx\right)$ for $a > 0$ and $b \in \mathbb{R}$. Which distribution is f?

Solution:

$$\begin{aligned} -\tfrac{1}{2}ax^2 + bx &= -\tfrac{1}{2}a\left(x^2 - \tfrac{2b}{a}x\right) \\ &= -\tfrac{1}{2}a\left(x - \tfrac{b}{a}\right)^2 + \tfrac{b^2}{2a} && \text{complete the square} \end{aligned}$$

The last term is free of x and is absorbed into the normalising constant, so

$$f(x) \propto \exp\left(-\frac{1}{2}\frac{(x - b/a)^2}{1/a}\right) \quad \Longrightarrow \quad f(x) = \mathcal{N}\left(x;\ \tfrac{b}{a},\ \tfrac{1}{a}\right),$$

where $\mathcal{N}(x; \mu, \sigma^2)$ denotes the density of a Gaussian with mean μ and variance σ^2 evaluated at x. A density is determined by its dependence on x alone, since the constant is fixed by $\int f = 1$.

(c) Now let f be a density on $\mathbb{R}^d$ with $f(x) \propto \exp\left(-\frac{1}{2}x^\top \Lambda x + b^\top x\right)$, where $b \in \mathbb{R}^d$ and Λ lies in the set of symmetric positive definite matrices,

$$\mathbf{S}^d_{++} = \left\{ M \in \mathbb{R}^{d\times d} : M = M^\top \text{ and } v^\top M v > 0 \text{ for all } v \in \mathbb{R}^d \setminus \{0\} \right\}.$$

Which distribution is f? Deduce the value of $\int_{\mathbb{R}^d} \exp\left(-\frac{1}{2}(x-m)^\top \Lambda (x-m)\right) dx$ for an arbitrary $m \in \mathbb{R}^d$.

Solution: Put $m = \Lambda^{-1}b$. Expanding the completed square,

$$\begin{aligned} -\tfrac{1}{2}(x-m)^\top \Lambda (x-m) &= -\tfrac{1}{2}x^\top \Lambda x + m^\top \Lambda x - \tfrac{1}{2}m^\top \Lambda m && \Lambda = \Lambda^\top \\ &= -\tfrac{1}{2}x^\top \Lambda x + b^\top x - \tfrac{1}{2}b^\top \Lambda^{-1} b, && m^\top \Lambda = b^\top \end{aligned}$$

which agrees with the given exponent up to a constant. Hence $f = \mathcal{N}(\Lambda^{-1}b, \Lambda^{-1})$, and Λ is the *precision* matrix. Comparing f with the Gaussian density gives the normalising constant,

$$\int_{\mathbb{R}^d} \exp\left(-\tfrac{1}{2}(x-m)^\top \Lambda (x-m)\right) dx = (2\pi)^{d/2} \det(\Lambda)^{-1/2} = \sqrt{\det(2\pi\Lambda^{-1})},$$

a constant with respect to m.

Exercise 1: True/False Question

(a) Let $X \sim \mathcal{N}(\mu, \Sigma)$ be a Gaussian random vector.

[☑] The marginal distribution of any subset of components of X is not Gaussian. ☐ True ■ **False**

Solution: As seen in the lecture, AX is Gaussian for any deterministic matrix A. Let $X = (X_1, \ldots, X_n)^\top$ and consider any subset of components $X_{i_1}, \ldots, X_{i_k}$. Choose the selection matrix

$$A = \begin{pmatrix} e_{i_1}^\top \\ \vdots \\ e_{i_k}^\top \end{pmatrix} \in \mathbb{R}^{k\times n},$$

where $e_i \in \mathbb{R}^n$ has a 1 in position i and zeros elsewhere. Then

$$AX = \begin{pmatrix} X_{i_1} \\ \vdots \\ X_{i_k} \end{pmatrix} \sim \mathcal{N}(A\mu,\, A\Sigma A^\top).$$

Thus, the joint marginal distribution of any subset of components of X is Gaussian.

[☑] Are linear combinations of jointly Gaussian random variables Gaussian? ■ **True** ☐ False

Solution: Let $X = (X_1, \ldots, X_n)^\top$ and consider a general linear combination

$$Y = a_1 X_1 + \cdots + a_n X_n,$$

where $a_1, \ldots, a_n \in \mathbb{R}$ are deterministic coefficients. Defining the vector

$$a = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{R}^n,$$

we can write

$$Y = a^\top X = \begin{pmatrix} a_1 & \cdots & a_n \end{pmatrix} \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}.$$

As seen in the lecture, AX is Gaussian for any deterministic matrix A. Taking $A = a^\top \in \mathbb{R}^{1\times n}$ gives

$$Y = a^\top X \sim \mathcal{N}\left(a^\top \mu,\, a^\top \Sigma a\right).$$

Thus, any linear combination of the components of X is Gaussian (possibly degenerate if its variance is zero).

(b) Let X, Y, Z be independent standard normal random variables.

[☑] The random variable T=$\frac{X+e^Z Y}{\sqrt{1+e^{2Z}}}$ is not Gaussian. ☐ True ■ **False**

Solution: In this question, it is difficult to calculate the distribution of $T = \frac{X+e^Z Y}{\sqrt{1+e^{2Z}}}$ directly since Z appears in both the numerator and the denominator inside non-linear exponential functions. The trick here is to first condition T on Z, and then integrate over the distribution of Z. As a clarification, we use lowercase p to denote the probability density function, and $f_Z(z)$ for the marginal density of Z. For $z \in \mathbb{R}$,

$$\begin{aligned} p(T \mid Z = z) &= p\left(\frac{X + e^Z Y}{\sqrt{1+e^{2Z}}} \;\middle|\; Z = z\right) \\ &= p\left(\frac{X + e^z Y}{\sqrt{1+e^{2z}}}\right) && X, Y \perp Z \\ &= \mathcal{N}(T; 0, 1) && \text{as } X + e^z Y \sim \mathcal{N}(0, 1+e^{2z}) \end{aligned}$$

which means that $p(T \mid Z = z)$ is always $\mathcal{N}(0,1)$, and is *independent from the value of* Z.

Using the law of total probability, we have

$$\begin{aligned} p(T) &= \int p(T \mid Z = z) f_Z(z)\, dz && \text{law of total probability} \\ &= \mathcal{N}(T; 0, 1) \int f_Z(z)\, dz && p(T \mid Z) \text{ does not depend on } Z \\ &= \mathcal{N}(T; 0, 1) && \textstyle\int f_Z(z)\, dz = 1 \end{aligned}$$

[☑] Let $\alpha \in \{0, \frac{\pi}{2}\}$ with $\mathbb{P}(\alpha = 0) = \mathbb{P}(\alpha = \frac{\pi}{2}) = \frac{1}{2}$ be independent of X and Y. Then $X\cos(\alpha) + Y\sin(\alpha) \sim \mathcal{N}(0,1)$. ■ **True** ☐ False

Solution: We will use the same trick as the one used in the previous question. Let $Z = X\cos\alpha +$

$Y \sin \alpha$, we have

$$\begin{aligned} p(Z|\alpha = a) &= p(X\cos\alpha + Y\sin\alpha|\alpha = a) \\ &= p(X\cos a + Y\sin a) && X, Y \perp \alpha \\ &= \mathcal{N}(Z; 0, \cos^2 a + \sin^2 a) && aX + bY \sim \mathcal{N}(0, a^2 + b^2) \\ &= \mathcal{N}(Z; 0, 1). \end{aligned}$$

By integrating over α one obtains the result. Note that the distribution of α is not important at all, only its independence from X and Y is important.

[☑] Let $(X, Y) \sim \mathcal{N}(0, \Sigma)$, where $\Sigma = \begin{pmatrix} 1 & \alpha \\ \alpha & 1 \end{pmatrix}$, and $\alpha \in \mathbb{R}, |\alpha| \leq 1$. Then random vectors $U = X + Y$ and $V = X - Y$ are independent if and only if $\alpha = 0$.

□ True ■ **False**

Solution: Jointly Gaussian random variables are independent if and only if they are uncorrelated. We can determine their independence by computing the covariance between U and V:

$$\begin{aligned} \text{Cov}(U, V) &= \text{Cov}(X + Y, X - Y) \\ &= \text{Cov}(X, X) - \text{Cov}(Y, Y) + \text{Cov}(Y, X) - \text{Cov}(X, Y) && \text{bilinearity of Cov} \end{aligned}$$

From the given covariance matrix Σ, we know the variances are $\text{Cov}(X, X) = 1$ and $\text{Cov}(Y, Y) = 1$, and the covariances are $\text{Cov}(X, Y) = \text{Cov}(Y, X) = \alpha$. Substituting these values into our equation gives:

$$\text{Cov}(U, V) = 1 - 1 + \alpha - \alpha = 0$$

Since the covariance is exactly 0 regardless of the value of α, the random vectors U and V are always independent. Therefore, their independence does not depend on α being equal to 0.

(c) [☑] Let $X_1, \ldots, X_n$ be mutually independent binary random variables. Completely specifying their joint distribution requires $\Theta(2^n)$ parameters.

□ True ■ **False**

Solution: The statement is false. Since the random variables are mutually independent, their joint distribution factorizes as

$$P(X_1 = x_1, \ldots, X_n = x_n) = \prod_{i=1}^{n} P(X_i = x_i).$$

Each binary random variable X_i is fully specified by a single parameter $p_i = P(X_i = 1)$, since $P(X_i = 0) = 1 - p_i$. Thus, only n parameters are needed, giving an $\mathcal{O}(n)$ representation.

Without independence, a general joint distribution has 2^n possible configurations and requires $2^n - 1$ free parameters, because the probabilities must sum to one. Independence allows all these probabilities to be computed from the n marginal parameters.

[☑] Let $Z \sim \mathcal{N}(\mu, \Sigma)$ be an n-dimensional multivariate Gaussian random variable. We can always find an invertible matrix A such that to completely specify the distribution of AZ, we need $\mathcal{O}(n)$ parameters in the worst case.i

■ **True** □ False

Solution: To completely specify a general n-dimensional Gaussian distribution $\mathcal{N}(\mu, \Sigma)$, we generally need n parameters for the mean vector μ and $\frac{n(n+1)}{2}$ parameters for the symmetric covariance matrix Σ (rather than n^2, due to symmetry). In the worst case, this requires $\mathcal{O}(n^2)$ parameters in total.

We want to find an invertible matrix A that simplifies the covariance matrix. We know from probability properties that for an affine transformation, the covariance matrix of the random variable AZ is given by $A\Sigma A^\top$.

Since Σ is a valid covariance matrix, it is symmetric and positive definite. By the Spectral Theorem, it can be diagonalized as:

$$\Sigma = PDP^\top$$

where P is an orthogonal matrix (meaning $P^\top P = PP^\top = I$) and D is a diagonal matrix containing the eigenvalues of Σ.

If we choose $A = P^\top$, then A is clearly invertible (its inverse is P). The transformed variable AZ will have the following covariance matrix:

$$\begin{aligned} \text{Cov}(AZ) &= A\Sigma A^\top \\ &= P^\top(PDP^\top)P && A = P^\top,\ \Sigma = PDP^\top \\ &= (P^\top P)D(P^\top P) = IDI = D && P^\top P = I \end{aligned}$$

Thus, $AZ \sim \mathcal{N}(P^\top\mu, D)$. To completely specify this new distribution, we now only need n parameters for the new mean vector $P^\top\mu$ and n parameters for the diagonal entries of D. This gives a total of $2n$ parameters, which satisfies the $\mathcal{O}(n)$ requirement.

Remark: We can reduce this even further. Because the eigenvalues of a positive definite matrix Σ are strictly positive, we can construct the inverse square root of the diagonal matrix, $D^{-1/2}$. If we instead choose $A = D^{-1/2}P^\top$, the covariance matrix of AZ reduces exactly to the identity matrix:

$$\begin{aligned} A\Sigma A^\top &= (D^{-1/2}P^\top)(PDP^\top)(PD^{-1/2}) \\ &= D^{-1/2}DD^{-1/2} = I && P^\top P = I \end{aligned}$$

In this scenario, we would only need n parameters to describe the new mean vector, while the covariance matrix requires 0 parameters to specify (since I is constant).

Exercise 2: Multivariate Gaussian Distribution

A vector-valued random variable $x \in \mathbb{R}^d$ is said to have a multivariate normal distribution with mean $\mu \in \mathbb{R}^d$ and covariance matrix $\Sigma \in \mathbf{S}^d_{++}$ if its pdf is:

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2}\det(\Sigma)^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$$

Prove the following facts, stated in the first lecture:

(a) Every marginal of a Gaussian vector is Gaussian (slide 51).

Solution: Consider $x = \begin{bmatrix} x_A \\ x_B \end{bmatrix}$, $\mu = \begin{bmatrix} \mu_A \\ \mu_B \end{bmatrix}$, and $\Sigma = \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{bmatrix}$, the joint distribution:

$$p(x) = p(x_A, x_B) = \frac{1}{Z} \exp\left(-\frac{1}{2} \begin{bmatrix} x_A - \mu_A \\ x_B - \mu_B \end{bmatrix}^T \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{bmatrix}^{-1} \begin{bmatrix} x_A - \mu_A \\ x_B - \mu_B \end{bmatrix} \right).$$

The following notations can ease the computation:

$$V = \begin{bmatrix} V_{AA} & V_{AB} \\ V_{BA} & V_{BB} \end{bmatrix} = \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{bmatrix}^{-1}, \quad \begin{bmatrix} x_A - \mu_A \\ x_B - \mu_B \end{bmatrix} = \begin{bmatrix} \Delta_A \\ \Delta_B \end{bmatrix}$$

$$\begin{aligned} p(x_A) =& \frac{1}{Z} \int_{x_B} \exp\left(-\frac{1}{2} \begin{bmatrix} \Delta_A \\ \Delta_B \end{bmatrix}^T \begin{bmatrix} V_{AA} & V_{AB} \\ V_{BA} & V_{BB} \end{bmatrix} \begin{bmatrix} \Delta_A \\ \Delta_B \end{bmatrix} \right) dx_B && \text{marginalise out } x_B \\ =& \frac{1}{Z} \exp\left(-\frac{1}{2} \left[\Delta_A^T (V_{AA} - V_{AB} V_{BB}^{-1} V_{BA}) \Delta_A \right] \right) \cdot \\ & \int_{x_B} \exp\left(-\frac{1}{2} \left[(\Delta_B + V_{BB}^{-1} V_{BA} \Delta_A)^T V_{BB} (\Delta_B + V_{BB}^{-1} V_{BA} \Delta_A) \right] \right) dx_B. && \text{complete the square} \end{aligned}$$

Note that the integrand is an unnormalized Gaussian pdf, so the value of the integral is $\sqrt{\det(2\pi V_{BB}^{-1})}$, which is a constant respect to x_A. We can therefore simplify it to

$$\begin{aligned} p(x_A) =& \frac{1}{Z_A} \exp\left(-\frac{1}{2} \left[\Delta_A^T (V_{AA} - V_{AB} V_{BB}^{-1} V_{BA}) \Delta_A \right] \right) = \\ =& \frac{1}{Z_A} \exp\left(-\frac{1}{2} \left[\Delta_A^T \Sigma_{AA}^{-1} \Delta_A \right] \right) && \Sigma_{AA}^{-1} = V_{AA} - V_{AB} V_{BB}^{-1} V_{BA} \end{aligned}$$

The last equality holds because:

$$\begin{bmatrix} V_{AA} & V_{AB} \\ V_{BA} & V_{BB} \end{bmatrix} = \begin{bmatrix} I & V_{AB} V_{BB}^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} V_{AA} - V_{AB} V_{BB}^{-1} V_{BA} & 0 \\ 0 & V_{BB} \end{bmatrix} \begin{bmatrix} I & 0 \\ V_{BB}^{-1} V_{BA} & I \end{bmatrix},$$

$$\begin{bmatrix} V_{AA} & V_{AB} \\ V_{BA} & V_{BB} \end{bmatrix}^{-1} = \begin{bmatrix} I & 0 \\ -V_{BB}^{-1} V_{BA} & I \end{bmatrix} \begin{bmatrix} (V_{AA} - V_{AB} V_{BB}^{-1} V_{BA})^{-1} & 0 \\ 0 & V_{BB}^{-1} \end{bmatrix} \begin{bmatrix} I & -V_{AB} V_{BB}^{-1} \\ 0 & I \end{bmatrix}.$$

The upper-left entry of $\begin{bmatrix} V_{AA} & V_{AB} \\ V_{BA} & V_{BB} \end{bmatrix}^{-1}$ is $(V_{AA} - V_{AB} V_{BB}^{-1} V_{BA})^{-1}$. Note that

$$\begin{bmatrix} V_{AA} & V_{AB} \\ V_{BA} & V_{BB} \end{bmatrix}^{-1} = \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{bmatrix},$$

hence

$$\Sigma_{AA} = (V_{AA} - V_{AB} V_{BB}^{-1} V_{BA})^{-1}.$$

Using similar arguments, we can also show that

$$V_{BB} = (\Sigma_{BB} - \Sigma_{BA} \Sigma_{AA}^{-1} \Sigma_{AB})^{-1}.$$

(b) Conditioning on a subset of variables of a joint Gaussian is Gaussian (slide 52).

Solution: Noting that $p(x_A)$ is a constant with respect to x_B, we have

$$
\begin{aligned}
p(x_B|x_A) =& \frac{p(x_A, x_B)}{p(x_A)} = \\
=& \frac{1}{Z'} \exp\left(-\frac{1}{2}\begin{bmatrix} x_A - \mu_A \\ x_B - \mu_B \end{bmatrix}^T \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{bmatrix}^{-1} \begin{bmatrix} x_A - \mu_A \\ x_B - \mu_B \end{bmatrix}\right) && p(x_A) \text{ absorbed into } Z' \\
=& \frac{1}{Z'} \exp\left(-\frac{1}{2}\left[\Delta_A^T (V_{AA} - V_{AB}V_{BB}^{-1}V_{BA})\Delta_A\right]\right) \cdot \\
& \cdot \exp\left(-\frac{1}{2}\left[(\Delta_B + V_{BB}^{-1}V_{BA}\Delta_A)^T V_{BB}(\Delta_B + V_{BB}^{-1}V_{BA}\Delta_A)\right]\right) && \text{complete the square} \\
=& \frac{1}{Z''} \exp\left(-\frac{1}{2}\left[(\Delta_B + V_{BB}^{-1}V_{BA}\Delta_A)^T V_{BB}(\Delta_B + V_{BB}^{-1}V_{BA}\Delta_A)\right]\right) && \text{1st factor free of } x_B
\end{aligned}
$$

$$
x_B|x_A \sim \mathcal{N}(\underbrace{\mu_B - V_{BB}^{-1}V_{BA}(x_A - \mu_A)}_{=\mu_{B|A}}; \underbrace{\Sigma_{BB} - \Sigma_{BA}\Sigma_{AA}^{-1}\Sigma_{AB}}_{=\Sigma_{B|A}=V_{BB}^{-1}})
$$

Recall the following fact about characteristic functions, which will help you to solve the next questions: For a random vector $X \in \mathbb{R}^d$, define its characteristic function φ_X as

$$\varphi_X(t) = \mathbb{E}[\exp(i\, t^\top X)], \quad \text{for all } t \in \mathbb{R}^d.$$

The characteristic function completely identifies a distribution. For a multivariate Normal distribution $\mathcal{N}(\mu, \Sigma)$, the characteristic function can be computed explicitly:

$$\varphi(t) = \exp(i\, t^\top \mu - \frac{1}{2} t^\top \Sigma t).$$

(c) [☑] Let $X = (X_1, \ldots, X_d)$ be a d-dimensional standard Gaussian random vector, that is, $X \sim \mathcal{N}_d(0, I)$. Define $Y = AX + \mu$, where A is a $d \times d$ matrix and $\mu \in \mathbb{R}^d$. What is the distribution of Y?

Solution: Let us compute the characteristic function of Y. Define $s = A^\top t$. We have

$$
\begin{aligned}
\varphi_Y(t) &= \mathbb{E}[\exp(it^\top Y)] && \\
&= \mathbb{E}[\exp(it^\top AX) \cdot \exp(it^\top \mu)] && Y = AX + \mu \\
&= \mathbb{E}[\exp(is^\top X)] \cdot \exp(it^\top \mu) && s = A^\top t \\
&= \varphi_X(s) \cdot \exp(it^\top \mu) && \text{def. of } \varphi_X \\
&= \exp(-\frac{1}{2} s^\top s + it^\top \mu) && X \sim \mathcal{N}(0, I) \\
&= \exp(it^\top \mu - -\frac{1}{2} t^\top AA^\top t), && s^\top s = t^\top AA^\top t
\end{aligned}
$$

which means that $Y \sim \mathcal{N}(\mu, AA^\top)$.

(d) [☑] If B is an $r \times d$ matrix, what is the distribution of BY?

Solution: With the same argument as the previous question, one gets $BY \sim \mathcal{N}(B\mu, BAA^\top B^\top)$.

(e) [☑] Let X, Y be i.i.d. $\mathcal{N}(0,1)$ random variables, then $X+Y$ and $X-Y$ are independent. ■ **True** □ False

Solution: The answer is true. Write $Z = (X,Y)^\top \sim \mathcal{N}\left(\begin{bmatrix}0\\0\end{bmatrix}\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}\right)$. Define $A = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}$, then $AZ = \begin{bmatrix}X+Y\\X-Y\end{bmatrix}$ and we hope to find the (joint) distribution of AZ.
Using the formulas from previous questions, we know that

$$AZ \sim \mathcal{N}(A0, AIA^\top) = \mathcal{N}\left(\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}2 & 0\\0 & 2\end{bmatrix}\right). \qquad \text{by (c) with } \mu = 0,\ \Sigma = I$$

This implies that $X+Y$ and $X-Y$ are uncorrelated. Remember that Gaussian random variables are uncorrelated if and only if they are independent. Hence $X+Y$ and $X-Y$ are independent.

(f) [☑] Let $X = (X_1, X_2)$ be a bivariate Normal random variable with mean $\mu = (1,1)$ and covariance matrix $\Sigma = \left(\begin{smallmatrix}5 & 1\\1 & 4\end{smallmatrix}\right)$. What is the mean and the variance of the conditional distribution of $Y = X_1 + X_2$ given $Z = X_1 - X_2 = 0$?

Solution: Since a linear transformation of a Gaussian vector is Gaussian,

$$\begin{bmatrix}Y\\Z\end{bmatrix} = AX, \qquad A = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix},$$

is jointly Gaussian with mean and covariance matrix

$$A\mu = \begin{bmatrix}2\\0\end{bmatrix}, \qquad A\Sigma A^\top = \begin{bmatrix}11 & 1\\1 & 7\end{bmatrix}.$$

Using the conditional density formula and omitting factors independent of y, we obtain

$$\begin{aligned} f_{Y|Z}(y \mid 0) &= \frac{f_{Y,Z}(y,0)}{f_Z(0)} \\ &\propto \exp\left(-\frac{1}{2}\begin{bmatrix}y-2\\0\end{bmatrix}^\top \frac{1}{76}\begin{bmatrix}7 & -1\\-1 & 11\end{bmatrix}\begin{bmatrix}y-2\\0\end{bmatrix}\right) \\ &= \exp\left(-\frac{(y-2)^2}{2(76/7)}\right). \end{aligned}$$

Thus,

$$Y \mid Z = 0 \sim \mathcal{N}\left(2, \frac{76}{7}\right),$$

so the conditional mean is 2 and the conditional variance is $\frac{76}{7}$.

(g) We consider a dataset $\{(x_i, y_i)\}_{i=1}^n$ of size n, where $x_i \in \mathbb{R}^d$ denotes the feature vector and $y_i \in \mathbb{R}$ denotes the label of the i-th data point. Let ϵ_i be i.i.d. samples from the Gaussian distribution $\mathcal{N}(0, \lambda^{-1})$ for a given $\lambda > 0$. We collect the labels in a vector $\mathbf{y} \in \mathbb{R}^n$, the features in a matrix $X \in \mathbb{R}^{n\times d}$, and the noise in a vector $\epsilon \in \mathbb{R}^n$. The labels are generated according to

$$y = Xw + \epsilon.$$

To perform Bayesian Linear Regression, we consider the prior distribution over the parameter vector w to be $\mathcal{N}(\mu, \lambda^{-1} I_d)$, where I_d denotes the d-dimensional identity matrix and $\mu \in \mathbb{R}^d$ is a hyperparameter.

[☑] The following statement holds for the conditional covariance matrix $\Sigma_y = \mathrm{Var}(y \mid X, \mu, \lambda) = \lambda^{-1} XX^\top + I_n$ □ True ■ **False**

Solution: Let $f = Xw$. Since $w \mid \mu, \lambda \sim \mathcal{N}(\mu, \lambda^{-1} I_d)$, we see from lecture 1, slide 52 that

$$f \mid X, \mu, \lambda \sim \mathcal{N}(X\mu, \lambda^{-1} XX^\top). \qquad \text{Cov}(Xw) = X(\lambda^{-1} I_d)X^\top$$

Thus,

$$y \mid X, \mu, \lambda \sim \mathcal{N}(X\mu, \lambda^{-1} XX^\top + \lambda^{-1} I_n). \qquad \epsilon \perp f,\ \epsilon \sim \mathcal{N}(0, \lambda^{-1} I_n)$$

[☑] Maximum likelihood estimate $\hat{\mu}_1$ of the hyperparameter μ is equal to $\mu_1 = \left(X^\top \Sigma_y^{-1} X\right)^{-1} X^\top \Sigma_y^{-1} y$. ■ **True** □ False

Solution: We have

$$\hat{\mu}_1 = \arg\max_\mu p(y \mid X, \mu, \lambda) = \arg\max_\mu \log \mathcal{N}(X\mu, \Sigma_y). \qquad \log \text{ is monotone}$$

We can simplify

$$\begin{aligned} -\log \mathcal{N}(X\mu, \Sigma_y) &= \frac{1}{2}(y - X\mu)^\top \Sigma_y^{-1}(y - X\mu) + \text{const} \\ &= \frac{1}{2}\mu^\top X^\top \Sigma_y^{-1} X\mu - \mu^\top X^\top \Sigma_y^{-1} y + \text{const.} \qquad \text{drop terms free of } \mu \end{aligned}$$

(Note that $\mu^\top X^\top \Sigma_y^{-1} y$ and $y^\top \Sigma_y^{-1} X\mu$ are both scalars so :

$$\mu^\top X^\top \Sigma_y^{-1} y = (\mu^\top X^\top \Sigma_y^{-1} y)^\top = y^\top \Sigma_y^{-1} X\mu$$

and Σ_y^{-1} is a symmetric matrix).
Taking the gradient with respect to μ, we obtain

$$\nabla_\mu \left(\frac{1}{2}\mu^\top X^\top \Sigma_y^{-1} X\mu - y^\top \Sigma_y^{-1} X\mu + \text{const} \right) = X^\top \Sigma_y^{-1} X\mu - X^\top \Sigma_y^{-1} y.$$

This is zero iff

$$\mu = \left(X^\top \Sigma_y^{-1} X\right)^{-1} X^\top \Sigma_y^{-1} y.$$

Exercise 3: Bayesian Inference with Conjugate Distributions

A company wants us to model the average lifetime of their newest LED lightbulbs. For this they observed the lifetime of n bulbs, giving us the dataset $\{x_1, \ldots, x_n\}$ with $x_i \in \mathbb{R}^+$. Due to LED bulbs having a constant failure rate, we assume $x_i \overset{\text{i.i.d}}{\sim} \text{Exp}(\lambda)$. Furthermore, we use a gamma distribution as the prior for the rate parameter λ, i.e., $\lambda \sim \text{Gamma}(\alpha, \beta), \alpha, \beta > 0$. The pdfs of $S \sim \text{Exp}(\lambda)$ and $T \sim \text{Gamma}(\alpha, \beta)$ are, respectively,

$$\begin{aligned} p_S(s) &= \lambda e^{-\lambda s}, & s \geq 0 \\ p_T(t) &= \frac{t^{\alpha-1} e^{-\beta t} \beta^\alpha}{\Gamma(\alpha)}, & t \geq 0, \end{aligned}$$

where $\Gamma(\cdot)$ is the gamma function. Let $\hat{\lambda}_{\text{MAP},n}$ be the MAP estimate of λ using the model defined above. Let $\hat{\lambda}_{\text{MLE},n}$ be the MLE estimate.

(a) [☑] What is the likelihood $p(x_{1:n} \mid \lambda)$?

Solution: We compute:

$$p\left(x_{1:n} \mid \lambda\right) = \prod_{i=1}^{n} p\left(x_i \mid \lambda\right) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i}.$$

The first equality uses that the x_i are conditionally independent given λ, so the joint density factorises over the data points. The second equality uses that they are identically distributed, so every factor is the same exponential density $p(x_i \mid \lambda) = \lambda e^{-\lambda x_i}$.

(b) [☑] What is the posterior distribution of the rate parameter $p\left(\lambda \mid x_{1:n}\right)$?

Solution: We compute:

$$\begin{aligned} p\left(\lambda \mid x_{1:n}\right) &\propto p\left(\lambda\right) p\left(x_{1:n} \mid \lambda\right) = p\left(\lambda\right) \prod_{i=1}^{n} p\left(x_i \mid \lambda\right) \\ &\propto \lambda^{\alpha-1} e^{-\beta\lambda} \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i} \\ &= \lambda^{(\alpha+n)-1} e^{-\lambda\left(\beta + \sum_{i=1}^{n} x_i\right)}. \end{aligned}$$

In the second line we dropped the factor $\beta^\alpha / \Gamma(\alpha)$ of the prior, since it does not depend on λ and is absorbed into the normalising constant of the posterior. Comparing the last line with the Gamma density, the posterior distribution is Gamma $(\alpha + n, \beta + \sum_{i=1}^{n} x_i)$.

(c) [☑] What is the MAP estimate of λ, i.e., $\hat{\lambda}_{\mathrm{MAP},n}$?

Solution: We first compute the mode of the Gamma(α, β) distribution, i.e., $t^* = \arg\max_t p_T(t)$ for $T \sim \text{Gamma}(\alpha, \beta)$.
By setting the derivative of the pdf of Gamma(α, β) to 0 we get

$$\begin{aligned} & 0 \overset{!}{=} \frac{\partial p_T(t)}{\partial t} \\ \Leftrightarrow \quad & 0 = \frac{\beta^\alpha}{\Gamma(\alpha)} \left((\alpha-1) t^{\alpha-2} e^{-\beta t} - \beta t^{\alpha-1} e^{-\beta t} \right) \qquad (1) \\ \Leftrightarrow \quad & 0 = \frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-2} e^{-\beta t} \left((\alpha-1) - \beta t \right) \\ \Leftrightarrow \quad & 0 = (\alpha-1) - \beta t \quad \text{or} \quad t = 0 \\ \Leftrightarrow \quad & t_1 = \frac{\alpha-1}{\beta}, \quad t_2 = 0 \end{aligned}$$

First we analyze the properties of t_2. Note that for $\alpha > 1$ we have $p_T(0) = 0$ and by definition of a pdf, we have $\int_0^\infty p_T(t)dt = 1$. Therefore, t_2 cannot be the mode of the pdf for $\alpha > 1$. For $0 < \alpha < 1$ $p_T(t)$ is a decreasing function (as both $t^{\alpha-1}$ and $e^{-\beta t}$ are decreasing functions), this implies that 0 is the mode of the distribution, note that $\lim_{t \to 0^+} p_T(t) = \infty$. Finally, for $\alpha = 1$ we have $p_T(t) = \beta e^{-\beta t}$ on $[0, \infty)$, using $\Gamma(1) = 1$. This is strictly decreasing, so $t_2 = 0$ is the mode, with $p_T(0) = \beta$. To classify the stationary point t_1 as a maximum rather than a minimum, we check the sign of the second derivative evaluated at t_1. We compute the second derivative,

$$\begin{aligned} \frac{\partial^2 p_T(t)}{\partial t^2} &= \frac{\beta^\alpha}{\Gamma(\alpha)} \Big((\alpha-1)(\alpha-2) t^{\alpha-3} e^{-\beta t} - \beta(\alpha-1) t^{\alpha-2} e^{-\beta t} \\ &\qquad - \beta(\alpha-1) t^{\alpha-2} e^{-\beta t} + \beta^2 t^{\alpha-1} e^{-\beta t} \Big) \\ &= C \left((\alpha-1)(\alpha-2) - 2\beta(\alpha-1)t + \beta^2 t^2 \right), \end{aligned}$$

where we applied the product rule twice to equation (1) and $C = \frac{\beta^\alpha}{\Gamma(\alpha)} e^{-\beta t} t^{\alpha-3}$ is a positive constant for $t \in (0, \infty)$. We now evaluate the second derivative at t_1,

$$\begin{aligned}
\left.\frac{\partial^2 p_T(t)}{\partial t^2}\right|_{t=t_1} &= C\left((\alpha-1)(\alpha-2) - 2\beta(\alpha-1)\frac{\alpha-1}{\beta} + \beta^2\left(\frac{\alpha-1}{\beta}\right)^2\right) \\
&= C\left((\alpha-1)(\alpha-2) - 2(\alpha-1)(\alpha-1) + (\alpha-1)^2\right) \\
&= C\,(\alpha-1)\left((\alpha-2) - (\alpha-1)\right) = C(1-\alpha)
\end{aligned}$$

As C is a positive constant we have,

$$\begin{aligned}
&\left.\frac{\partial^2 p_T(t)}{\partial t^2}\right|_{t=t_1} < 0 \\
\Leftrightarrow \quad &\alpha > 1
\end{aligned}$$

The mode of the Gamma distribution is

$$t^* = \begin{cases} 0, & 0 < \alpha \leq 1 \\ \frac{\alpha-1}{\beta}, & \alpha > 1 \end{cases}$$

As $\alpha + n > 1$, the MAP estimate is given by

$$\begin{aligned}
\hat{\lambda}_{\mathrm{MAP},n} &= \arg\max_{\lambda} p(\lambda \mid x_{1:n}) \\
&= \text{mode of Gamma}\left(\alpha + n, \beta + \sum_{i=1}^{n} x_i\right) \\
&= \frac{\alpha + n - 1}{\beta + \sum_{i=1}^{n} x_i}.
\end{aligned}$$

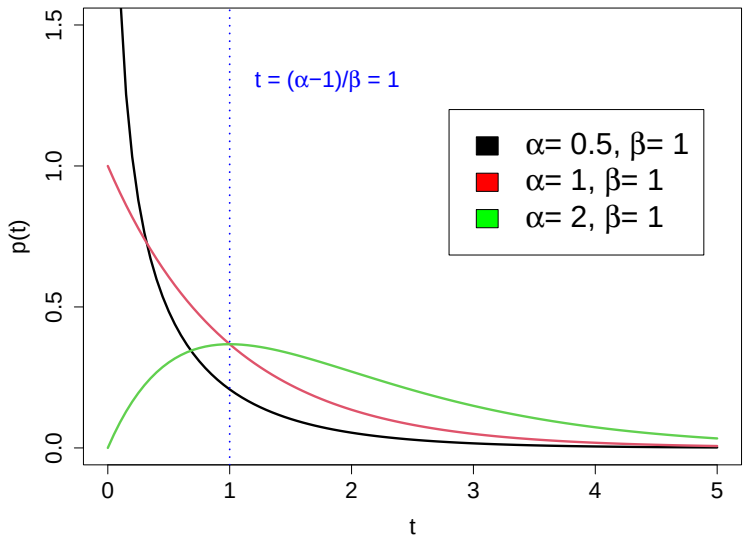


Figure 1: Pdf of Gamma distribution with different α.

(d) [☑] If n is sufficiently large, $\hat{\lambda}_{\mathrm{MAP},n} \approx \hat{\lambda}_{\mathrm{MLE},n}$. ■ **True** □ False

Solution: The answer is true.
To compute $\hat{\lambda}_{\text{MLE},n}$, we compute the log-likelihood

$$\begin{aligned}\log p(x_{1:n} \mid \lambda) &= \sum_{i=1}^{n} \log p(x_i \mid \lambda) \\ &= \sum_{i=1}^{n} \log(\lambda) - \lambda x_i \\ &= n \log \lambda - \lambda \sum_{i=1}^{n} x_i.\end{aligned}$$

We next aim to maximize the log-likelihood.

$$\frac{\partial \log p(x_{1:n} \mid \lambda)}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i \stackrel{!}{=} 0.$$

Solving the equation above gives

$$\hat{\lambda}_{\text{MLE},n} = \frac{1}{\frac{\sum_{i=1}^{n} x_i}{n}} = \frac{1}{\overline{X}_n}.$$

Now let n grow. The x_i are i.i.d. with $\mathbb{E}[x_i] = 1/\lambda < \infty$, so the strong law of large numbers gives $\overline{X}_n \to 1/\lambda$ almost surely. This limit is strictly positive, so $1/\overline{X}_n$ converges as well, and

$$\begin{aligned}\lim_{n\to\infty} \hat{\lambda}_{\text{MLE},n} &= \lim_{n\to\infty} \frac{1}{\overline{X}_n} = \lambda \\ \lim_{n\to\infty} \hat{\lambda}_{\text{MAP},n} &= \lim_{n\to\infty} \frac{\alpha + n - 1}{\beta + \sum_{i=1}^{n} x_i} = \lim_{n\to\infty} \frac{\frac{\alpha-1}{n} + 1}{\frac{\beta}{n} + \overline{X}_n} = \frac{1}{1/\lambda} = \lambda\end{aligned}$$

Both estimators converge to the same limit λ, and that is what $\hat{\lambda}_{\text{MAP},n} \approx \hat{\lambda}_{\text{MLE},n}$ means for large n. The limit cannot be dropped earlier, because $\overline{X}_n$ itself depends on n.

> The same behaviour holds more generally, but not unconditionally. If the model is regular and the prior places positive density on a neighbourhood of the true parameter, then the likelihood contribution grows with n while the prior contribution does not, so the prior is washed out and the MAP estimate approaches the MLE. A prior that assigns zero density to the true parameter, or that is degenerate, keeps its influence no matter how much data is collected, and the two estimates need not agree.

Exercise 4: Online Bayesian Linear Regression

Consider Bayesian linear regression (BLR) with the model

$$w \sim \mathcal{N}(0, I_d), \qquad y_i = x_i^\top w + \varepsilon_i, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_n^2),$$

where $x_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}$, and the noise variance $\sigma_n^2 > 0$ is known. We condition on the feature vectors, treating them as fixed inputs, and assume that the observation noises are independent of w. Here, I_d denotes the $d \times d$ identity matrix. As seen in the lecture, this model has a Gaussian posterior. If X is the data matrix whose ith row is $x_i^\top$ and y is the vector of responses, then

$$p(w \mid X, y) = \mathcal{N}\Big(w; (X^\top X + \sigma_n^2 I_d)^{-1} X^\top y, (\sigma_n^{-2} X^\top X + I_d)^{-1}\Big).$$

Instead of having all observations available at once, suppose that one new observation (x_t, y_t) arrives at each round $t = 1, 2, \ldots$. Let

$$\mathcal{D}_t := \{(x_i, y_i)\}_{i=1}^t, \qquad p_t(w) := p(w \mid \mathcal{D}_t),$$

with $\mathcal{D}_0 = \varnothing$ and $p_0(w) = \mathcal{N}(w; 0, I_d)$. Our goal is to update the posterior after each observation and to compute the predictive distribution of a new noisy response $y^\star$ at a given feature vector $x^\star$. One approach is to construct the full data matrix $X^{(t)} \in \mathbb{R}^{t \times d}$ and response vector $y^{(t)} \in \mathbb{R}^t$ at every round, then recompute the posterior using the formula above. This gives the correct result, but storing and repeatedly processing all past observations becomes increasingly costly as t grows. Although the matrix to be inverted remains $d \times d$, recomputing its entries from the full dataset requires an increasing amount of work. The following questions first remove the need to retain past observations and then remove the need to recompute a matrix inverse at every round.

(a) Let $X = X^{(t)}$ be the data matrix whose ith row is $x_i^\top$, and let $y = y^{(t)} = (y_1, \ldots, y_t)^\top$. Prove that

$$X^\top X = \sum_{i=1}^t x_i x_i^\top.$$

Derive an analogous expression for $X^\top y$.

Solution: Since the rows of X are $x_i^\top$, block matrix multiplication gives

$$X^\top X = \begin{bmatrix} x_1 & \cdots & x_t \end{bmatrix} \begin{bmatrix} x_1^\top \\ \vdots \\ x_t^\top \end{bmatrix} = \sum_{i=1}^t x_i x_i^\top.$$

Similarly,

$$X^\top y = \begin{bmatrix} x_1 & \cdots & x_t \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_t \end{bmatrix} = \sum_{i=1}^t y_i x_i.$$

The first sum is a $d \times d$ matrix, and the second is a vector in $\mathbb{R}^d$.

(b) Design an algorithm that updates the posterior as observations arrive, using memory independent of t for fixed d. Give its initialization and update rules, explain how to obtain the posterior parameters, and justify the memory requirement. You may still compute a matrix inverse at each round.

Hint: Use part (a) to identify sums that can be updated one observation at a time.

Solution: As discussed in the lecture and recalled in the introduction,

$$p_t(w) = \mathcal{N}(w; \bar{\mu}_t, \bar{\Sigma}_t),$$

with

$$\bar{\mu}_t = \left({X^{(t)}}^\top X^{(t)} + \sigma_n^2 I_d \right)^{-1} {X^{(t)}}^\top y^{(t)},$$
$$\bar{\Sigma}_t = \left(\sigma_n^{-2} {X^{(t)}}^\top X^{(t)} + I_d \right)^{-1}.$$

Thus, we only need the two quantities ${X^{(t)}}^\top X^{(t)}$ and ${X^{(t)}}^\top y^{(t)}$ to compute the posterior.

By part (a),

$$ {X^{(t)}}^\top X^{(t)} = \sum_{i=1}^{t} x_i x_i^\top, \qquad {X^{(t)}}^\top y^{(t)} = \sum_{i=1}^{t} y_i x_i.$$

When the next observation (x_{t+1}, y_{t+1}) arrives, each sum gains just one term:

$$ {X^{(t+1)}}^\top X^{(t+1)} = {X^{(t)}}^\top X^{(t)} + x_{t+1} x_{t+1}^\top,$$
$$ {X^{(t+1)}}^\top y^{(t+1)} = {X^{(t)}}^\top y^{(t)} + y_{t+1} x_{t+1}.$$

There is therefore no need to recompute these sums from the previous observations.

More explicitly, store

$$S_t := {X^{(t)}}^\top X^{(t)}, \qquad b_t := {X^{(t)}}^\top y^{(t)}.$$

Initialize

$$S_0 = 0_{d \times d}, \qquad b_0 = 0_d,$$

and, for each new observation, update

$$S_{t+1} = S_t + x_{t+1} x_{t+1}^\top, \qquad b_{t+1} = b_t + y_{t+1} x_{t+1}.$$

The updated posterior parameters are then

$$\bar{\mu}_{t+1} = (S_{t+1} + \sigma_n^2 I_d)^{-1} b_{t+1}, \qquad \bar{\Sigma}_{t+1} = (I_d + \sigma_n^{-2} S_{t+1})^{-1}.$$

Since S_t is a $d \times d$ matrix and b_t is a vector in $\mathbb{R}^d$, storing them requires $O(d^2)$ memory, independently of t. Once an observation has been incorporated into these sums, it no longer needs to be stored.

Updating the sums takes $O(d^2)$ time per round. Computing the posterior parameters using a standard dense matrix inversion still costs $O(d^3)$ per round; the subsequent questions address this remaining cost.

(c) The Sherman–Morrison identity states that, for a symmetric invertible matrix $A \in \mathbb{R}^{d \times d}$ and a vector $x \in \mathbb{R}^d$ such that $1 + x^\top A^{-1} x \neq 0$,

$$(A + xx^\top)^{-1} = A^{-1} - \frac{(A^{-1}x)(A^{-1}x)^\top}{1 + x^\top A^{-1} x}.$$

Prove this identity by multiplying the proposed inverse by $A + xx^\top$.

Solution: Define

$$v := A^{-1}x, \qquad s := 1 + x^\top v, \qquad B := A^{-1} - \frac{vv^\top}{s}.$$

Since A is symmetric, A^{-1} is also symmetric, so

$$v^\top = x^\top A^{-1}, \qquad Av = x.$$

Moreover,

$$(A + xx^\top)v = x + x(x^\top v) = sx.$$

Consequently,

$$\begin{aligned}(A + xx^\top)B &= (A + xx^\top)A^{-1} - \frac{(A + xx^\top)vv^\top}{s} \\ &= I_d + xv^\top - \frac{sxv^\top}{s} \\ &= I_d.\end{aligned}$$

For square matrices, this establishes that B is the inverse of $A + xx^\top$, proving the identity.
In particular, if A is positive definite, then $x^\top A^{-1}x \geq 0$, so the denominator is strictly positive.

(d) Computing a matrix inverse at each round costs $O(d^3)$. Modify the algorithm from part (b) to update the posterior parameters in $O(d^2)$ time per round. Specify the quantities to store, their initialization and update rules, and justify the time and memory requirements.

Hint: Apply the identity from part (c) to update the inverse using the new observation.

Solution: Define

$$A_t := S_t + \sigma_n^2 I_d, \qquad M_t := A_t^{-1}.$$

Since $\sigma_n^2 > 0$, A_t is positive definite. By part (b),

$$A_{t+1} = A_t + x_{t+1}x_{t+1}^\top.$$

Applying part (c) therefore gives

$$M_{t+1} = M_t - \frac{(M_t x_{t+1})(M_t x_{t+1})^\top}{1 + x_{t+1}^\top M_t x_{t+1}}.$$

It suffices to store M_t and b_t, initialized as

$$M_0 = \sigma_n^{-2} I_d, \qquad b_0 = 0_d.$$

For each new observation (x_{t+1}, y_{t+1}), compute

$$\begin{aligned}v &= M_t x_{t+1}, \\ s &= 1 + x_{t+1}^\top v, \\ M_{t+1} &= M_t - \frac{vv^\top}{s}, \\ b_{t+1} &= b_t + y_{t+1}x_{t+1}.\end{aligned}$$

The denominator satisfies $s \geq 1$ because M_t is positive definite.
The posterior parameters are

$$\bar{\mu}_{t+1} = M_{t+1}b_{t+1}, \qquad \bar{\Sigma}_{t+1} = \sigma_n^2 M_{t+1}.$$

The covariance formula follows from

$$(\sigma_n^{-2}S_{t+1} + I_d)^{-1} = \sigma_n^2(S_{t+1} + \sigma_n^2 I_d)^{-1}.$$

The initialization gives

$$\bar{\mu}_0 = 0_d, \qquad \bar{\Sigma}_0 = I_d,$$

as required by the prior.

Computing v, forming and subtracting $vv^\top/s$, and computing the posterior mean each take $O(d^2)$ time. Computing s and updating b_t each take $O(d)$ time. Forming the covariance also takes $O(d^2)$ time.

Thus, the total time per round is $O(d^2)$, and storing M_t, b_t, and the required working quantities takes $O(d^2)$ memory, independently of t. There is no need to retain S_t or past observations in this implementation.

In the naive approach, forming $X^\top X$ from all t observations costs $O(td^2)$, forming $X^\top y$ costs $O(td)$, and a standard dense matrix inversion costs $O(d^3)$. Storing the data and working matrices requires $O(td + d^2)$ memory.

The three approaches therefore have the following costs:

Approach	Time per round	Memory
Recompute from all observations	$O(td^2 + d^3)$	$O(td + d^2)$
Update S_t, b_t; recompute the inverse	$O(d^3)$	$O(d^2)$
Update M_t, b_t	$O(d^2)$	$O(d^2)$

All three methods give the same posterior and predictive distributions in exact arithmetic. In practice, floating-point rounding may introduce numerical differences.

(e) Using the posterior after observing (x_{t+1}, y_{t+1}) in part (d), derive the predictive distribution of a new noisy response $y^\star$ at a given feature vector $x^\star \in \mathbb{R}^d$.

Express its mean and variance in terms of $x^\star$, $\bar{\mu}_{t+1}$, $\bar{\Sigma}_{t+1}$, and σ_n^2.

Solution: The predictive distribution is obtained by integrating over the posterior uncertainty in w:

$$p(y^\star \mid x^\star, \mathcal{D}_{t+1}) = \int p(y^\star \mid x^\star, w)\, p(w \mid \mathcal{D}_{t+1})\, dw.$$

Under the observation model,

$$y^\star = x^{\star\top} w + \varepsilon^\star, \qquad \varepsilon^\star \sim \mathcal{N}(0, \sigma_n^2),$$

where the new noise $\varepsilon^\star$ is independent of w and the observed data.

Since

$$w \mid \mathcal{D}_{t+1} \sim \mathcal{N}(\bar{\mu}_{t+1}, \bar{\Sigma}_{t+1}),$$

the linear transformation $x^{\star\top} w$ is Gaussian with mean $x^{\star\top}\bar{\mu}_{t+1}$ and variance $x^{\star\top}\bar{\Sigma}_{t+1}x^\star$.

Adding the independent Gaussian observation noise gives

$$p(y^\star \mid x^\star, \mathcal{D}_{t+1}) = \mathcal{N}\left(y^\star; x^{\star\top}\bar{\mu}_{t+1}, \sigma_n^2 + x^{\star\top}\bar{\Sigma}_{t+1}x^\star\right).$$

The predictive variance combines observation noise, σ_n^2, and uncertainty about the weights, $x^{\star\top}\bar{\Sigma}_{t+1}x^\star$.